

§ Divisibility For integers a, b and positive integer n , $n \mid a$ if and only if there exists an integer k such that

$$a = kn.$$

§ Coprime (Relatively Prime) Two integers a and b are **coprime** if $\gcd(a, b) = 1$.

§ Greatest Common Divisor (gcd) For integers a, b , not both zero, the greatest common divisor of a and b , denoted $\gcd(a, b)$, is the largest positive integer that divides both a and b .

1. Integers a, b with $\gcd(a, b) = 3$ and $a + b = 65$.

Answer: If $\gcd(a, b) = 3$, then $3 \mid a$ and $3 \mid b$, so $3 \mid (a + b)$. But $3 \nmid 65$. Contradiction. **No such integers.**

□

Linear Congruences

§ Congruence For integers a, b and positive integer n , $n \mid (a - b)$ if

$$a \equiv b \pmod{n},$$

where a and b have the same **residue (remainder)** when divided by n . It follows that:

- If $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$, then $a + c \equiv b + d \pmod{n}$.
- If $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$, then $ac \equiv bd \pmod{n}$.
- If $a \equiv b \pmod{n}$ and $d \mid n$, then $a \equiv b \pmod{d}$.

Solvability: Let $d = \gcd(a, n)$. The congruence $ax \equiv b \pmod{n}$ has d distinct solutions modulo n iff

$$d \mid b$$

with the least positive solution x where $(a/d)x \equiv (b/d) \pmod{n/d}$.

1. $84x \equiv 5 \pmod{49}$.

Answer: $\gcd(84, 49) = 7$. Since $7 \nmid 5$, by theorem 1 there is **no solution**.

□

2. $12x \equiv 16 \pmod{32}$.

Answer: $\gcd(12, 32) = 4$ and $4 \mid 16$. Divide by 4: $3x \equiv 4 \pmod{8}$. Since $\gcd(3, 8) = 1$ and $3^{-1} \equiv 3 \pmod{8}$, we get $x \equiv 3 \cdot 4 = 12 \equiv 4 \pmod{8}$. The least positive solution is $\boxed{4}$. (All solutions mod 32: 4, 12, 20, 28.)

□

Multiplicative Inverses in Congruences

§ Multiplicative Inverse Modulo n An integer a has a multiplicative inverse modulo n if and only if $\gcd(a, n) = 1$.

1. Find the inverse of 7 modulo 24 (i.e., solve $7x \equiv 1 \pmod{24}$).

Answer: Since $\gcd(7, 24) = 1$, an inverse exists. Use the extended Euclidean algorithm:

$$\underline{24} = 3 \cdot \underline{7} + 3$$

$$7 = 2 \cdot 3 + \underline{1}$$

$$3 = 3 \cdot 1 + 0$$

Back-substitute, $(7) \cdot \underline{7} + (-2) \cdot \underline{24} = \underline{1}$. That is,

$$\underbrace{7}_x \cdot \underline{7} = 2 \cdot \underline{24} + \underline{1}.$$

Thus, $7^{-1} \equiv 7 \pmod{24}$. Least positive solution: $\boxed{7}$.

□

2. Find the inverse of 831 modulo 402 (i.e., solve $831x \equiv 1 \pmod{402}$) or show that no such inverse exists.

Answer: By the extended Euclidean algorithm:

$$\underline{831} = 2 \cdot \underline{402} + 27$$

$$402 = 14 \cdot 27 + 24$$

$$27 = 1 \cdot 24 + \underline{3}$$

$$24 = 8 \cdot 3 + 0$$

Back-substitute, $(15) \cdot \underline{831} + (-31) \cdot \underline{402} = \underline{3}$. Since $\gcd(831, 402) = 3 \neq 1$, **no inverse exists**. But still,

$$\underbrace{15}_x \cdot \underline{831} = (31) \cdot \underline{402} + \underline{3}.$$

That is, $831x \equiv 3 \pmod{402}$ has solutions. Dividing by 3, $277x \equiv 1 \pmod{134}$. Since $\gcd(277, 134) = 1$, an inverse exists. By the extended Euclidean algorithm,

$$\underline{277} = 2 \cdot \underline{134} + 9$$

$$134 = 14 \cdot 9 + 8$$

$$9 = 1 \cdot 8 + \underline{1}$$

$$8 = 8 \cdot 1 + 0$$

Back-substitute, $(15) \cdot \underline{277} + (-31) \cdot \underline{134} = \underline{1}$. That is,

$$\underbrace{15}_x \cdot \underline{277} = 31 \cdot \underline{134} + \underline{1}.$$

Thus, $277^{-1} \equiv 15 \pmod{134}$. Least positive solution: $\boxed{15}$. (All solutions mod 402: 15, 149, 283.)

□

Systems of Linear Congruences

§ Chinese Remainder Theorem Let $m_1, m_2, \dots, m_n$ be pairwise coprime positive integers, and let $b_1, b_2, \dots, b_n \in \mathbb{Z}$. Then the system of congruences

$$\begin{cases} x \equiv b_1 \pmod{m_1} \\ x \equiv b_2 \pmod{m_2} \\ \vdots \\ x \equiv b_n \pmod{m_n} \end{cases}$$

has a solution $x \in \mathbb{Z}$, and all solutions are congruent modulo $M := m_1 m_2 \cdots m_n$.

1. System $x \equiv 3 \pmod{7}$, $x \equiv 4 \pmod{6}$.

Answer: $\gcd(7, 6) = 1$. Let $x = 3 + 7k$. Then $3 + 7k \equiv 4 \pmod{6} \Rightarrow 7k \equiv 1 \pmod{6}$. Since $7 \equiv 1 \pmod{6}$, $k \equiv 1 \pmod{6}$. Take $k = 1$ gives $x = 10$. Hence $x \equiv 10 \pmod{42}$. Least positive: 10.

□

- 2.

$$\begin{aligned} x &\equiv 3 \pmod{4} \\ x &\equiv 2 \pmod{5} \end{aligned}$$

Answer: Since $\gcd(4, 5) = 1$, there is a unique solution modulo 20. Let $x = 4k + 3$. Then $4k + 3 \equiv 2 \pmod{5}$ gives $4k \equiv 4 \pmod{5}$, so $k \equiv 1 \pmod{5}$ and $x = 20m + 7$.

Therefore, by the Chinese Remainder Theorem, the least non-negative solution is $x = 7$.

□

- 3.

$$\begin{aligned} 3x &\equiv 2 \pmod{4} \\ 4x &\equiv 1 \pmod{5} \\ 6x &\equiv 3 \pmod{9} \end{aligned}$$

Answer: Reduce each congruence. Since $3^{-1} \equiv 3 \pmod{4}$, $x \equiv 2 \cdot 3 \equiv 2 \pmod{4}$. Since $4^{-1} \equiv 4 \pmod{5}$, $x \equiv 1 \cdot 4 \equiv 4 \pmod{5}$. That is,

$$\begin{aligned} x &\equiv 2 \pmod{4} \\ x &\equiv 4 \pmod{5} \\ x &\equiv 2 \pmod{3}. \end{aligned}$$

Let $x = 4k + 2$. Then $4k \equiv 0 \pmod{3}$ so $k \equiv 0 \pmod{3}$, giving $x \equiv 2 \pmod{12}$. With $x = 12k + 2$ and $x \equiv 4 \pmod{5}$, we obtain $12k \equiv 2 \pmod{5}$, so $k \equiv 1 \pmod{5}$ and $x = 60n + 14$.

Therefore, by the Chinese Remainder Theorem, the least non-negative solution is $x = 14$.

□

Reduced Residue Systems

§ Reduced Residue System Modulo n The symbol $(\mathbb{Z}/n\mathbb{Z})^\times$ denotes the **reduced residue system modulo n** , also called the **group of units modulo n** . It consists of all congruence classes $[a]$ in $\mathbb{Z}/n\mathbb{Z}$ such that $\gcd(a, n) = 1$:

$$(\mathbb{Z}/n\mathbb{Z})^\times = \{[a] \in \mathbb{Z}/n\mathbb{Z} \mid \gcd(a, n) = 1\}.$$

where

$$[a] = \{x \in \mathbb{Z} \mid x \equiv a \pmod{n}\}$$

is the equivalence class of integers congruent to a modulo n . It follows that

- **Closure:** If $[a], [b] \in (\mathbb{Z}/n\mathbb{Z})^\times$, then $[a][b] \in (\mathbb{Z}/n\mathbb{Z})^\times$.
- **Identity:** $[1]$ is the multiplicative identity.
- **Inverse:** Each $[a]$ has an inverse $[b]$ such that $[a][b] = [1]$.
- **Associativity:** Multiplication modulo n is associative.

1. List the **least non-negative reduced residue system** modulo 10.

Answer: A reduced residue system modulo 10 is a set of $\varphi(10) = 4$ integers that are coprime to 10. That is, $\gcd(1, 10) = 1$, $\gcd(3, 10) = 1$, $\gcd(7, 10) = 1$, and $\gcd(9, 10) = 1$. Thus, the units (least non-negative) reduced residue system is $\{1, 3, 7, 9\}$.

□

2. Is $\{-15, 66, -75, 11\}$ a reduced residue system modulo 8?

Answer: Reduce mod 8: $-15 \equiv 1$, $66 \equiv 2$, $-75 \equiv 5$, $11 \equiv 3$. Residues $\{1, 2, 5, 3\}$. But 2 is not coprime to 8 ($\gcd = 2$). Thus the set is **not** a reduced residue system modulo 8.

□

3. Reduced residue system modulo 7 consisting only of odd integers.

Answer: One choice: $\{1, 3, 5, 9, 11, 13\}$. Modulo 7 these reduce to $\{1, 3, 5, 2, 4, 6\}$, i.e. all units.

□

Factorials in Congruences

§ Wilson's Theorem For prime p , $(p-1)! \equiv -1 \pmod{p}$.

1. For odd prime p , prove $2(p-3)! \equiv -1 \pmod{p}$.

Proof: By Wilson's theorem, $(p-1)! \equiv -1 \pmod{p}$. But $(p-1)! = (p-1)(p-2)(p-3)! \equiv (-1)(-2)(p-3)! \equiv 2(p-3)! \pmod{p}$. Thus $2(p-3)! \equiv -1 \pmod{p}$.

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Multiplicative Arithmetic Functions

§ Multiplicativity A function $f : \mathbb{Z}_{>0} \rightarrow \mathbb{C}$ is **multiplicative** if

$$f(mn) = f(m)f(n) \text{ for any coprime positive integers } m, n.$$

completely multiplicative if

$$f(mn) = f(m)f(n) \text{ for all positive integers } m, n.$$

§ Chinese Remainder Theorem (pair): If $\gcd(m, n) = 1$ (or coprime), then

$$(\mathbb{Z}/mn\mathbb{Z})^\times \cong (\mathbb{Z}/m\mathbb{Z})^\times \times (\mathbb{Z}/n\mathbb{Z})^\times.$$

Taking cardinalities gives multiplicativity of φ on coprime arguments.

§ Euler's Totient Function $\varphi(n)$ The function $\varphi : \mathbb{Z}_{>0} \rightarrow \mathbb{Z}_{>0}$ is defined by

$$\varphi(n) = |(\mathbb{Z}/n\mathbb{Z})^\times| = |\{1 \leq k \leq n \mid (k, n) = 1\}|,$$

i.e. $\varphi(n)$ is the number of integers in $\{1, 2, \dots, n\}$ that are coprime to n .

Note: If n is prime, then $\phi(n) = n - 1$.

1. Prove $\varphi(p^k) = p^k - p^{k-1} = p^{k-1}(p - 1)$ for prime p and integer $k \geq 1$.

Proof: The integers in $\{1, 2, \dots, p^k\}$ that are not coprime to p^k are exactly the multiples of p : $1p, 2p, 3p, \dots, p^{k-1}p$. There are p^{k-1} such multiples. Thus,

$$\varphi(p^k) = p^k - p^{k-1} = p^{k-1}(p - 1).$$

■

2. Prove that φ is multiplicative but not completely multiplicative.

Proof: If $\gcd(m, n) = 1$ then by the Chinese Remainder Theorem,

$$|(\mathbb{Z}/mn\mathbb{Z})^\times| = |(\mathbb{Z}/m\mathbb{Z})^\times| \cdot |(\mathbb{Z}/n\mathbb{Z})^\times|,$$

so $\varphi(mn) = \varphi(m)\varphi(n)$. Thus, φ is **multiplicative**.

However, for prime p , $\varphi(p^k) = p^k - p^{k-1} = p^{k-1}(p - 1)$. Let $k = 2$. Then $\varphi(p^2) = p(p - 1)$ but $\varphi(p)\varphi(p) = (p - 1)(p - 1) = (p - 1)^2 \neq p(p - 1)$ for $p > 2$. Thus, φ is **not completely multiplicative**.

■

Exponentiation in Congruences

§ Euler's theorem If $\gcd(a, m) = 1$ then

$$a^{\varphi(m)} \equiv 1 \pmod{m}.$$

Proof: Let $R = \{r_1, r_2, \dots, r_{\phi(n)}\}$ be the reduced residue system modulo n . Since $\gcd(a, n) = 1$, the products $ar_1, ar_2, \dots, ar_{\phi(n)}$ form another reduced residue system modulo n . Thus,

$$ar_1 \cdot ar_2 \cdots ar_{\phi(n)} \equiv r_1 r_2 \cdots r_{\phi(n)} \pmod{n}.$$

That is,

$$a^{\phi(n)}(r_1 r_2 \cdots r_{\phi(n)}) \equiv r_1 r_2 \cdots r_{\phi(n)} \pmod{n}.$$

Since $r_1 r_2 \cdots r_{\phi(n)}$ are coprimes of n ,

$$a^{\phi(n)} \equiv 1 \pmod{n}.$$

■

Corollary: Fermat's little theorem: If $\gcd(a, m) = 1$ and m is prime then

$$a^{m-1} \equiv 1 \pmod{m}.$$

1. If $\gcd(a, m) = 1$, show $a^{\varphi(m)-1}$ is the inverse of a modulo m

Proof: By Euler's theorem $a^{\varphi(m)} \equiv 1 \pmod{m}$. Since $a^{\varphi(m)} = a \cdot a^{\varphi(m)-1}$,

$$a \cdot a^{\varphi(m)-1} \equiv a^{\varphi(m)} \equiv 1 \pmod{m},$$

so $a^{\varphi(m)-1}$ is an inverse of a modulo m . (Cited: Euler's theorem.)

■

2. Find the least positive solution x to the congruence $x \equiv 20^{110} \pmod{17}$.

Answer: $x \equiv 20^{110} \pmod{17}$. Reduce base: $20 \equiv 3 \pmod{17}$. Since 17 is prime, by Fermat, $3^{16} \equiv 1 \pmod{17}$, and $110 \equiv 14 \pmod{16}$. Thus

$$3^{110} \equiv 3^{14} = (3^8)(3^4)(3^2) \equiv 16 \cdot 13 \cdot 9 \equiv 4 \cdot 9 \equiv 36 \equiv 2 \pmod{17}.$$

So least positive solution: $\boxed{2}$.

□

3. For $x \equiv 38^{110} \pmod{21}$.

Answer: $x \equiv 38^{110} \pmod{21}$. Reduce base: $38 \equiv 17 \pmod{21}$. By Euler theorem and Multiplicativity of φ , $\varphi(21) = \varphi(3)\varphi(7) = 2 \cdot 6 = 12$,

$$17^{\varphi(21)} \equiv 17^{12} \equiv 1 \pmod{21}.$$

Thus,

$$38^{110} \equiv 17^{110} \equiv 17^2 \equiv 289 \equiv 16 \pmod{21}.$$

So least positive solution: $\boxed{16}$.

□

1. Prove Euler's theorem.

Proof: Let $m > 1$ and let $\{r_1, \dots, r_{\varphi(m)}\}$ be a reduced residue system modulo m . Since $\gcd(a, m) = 1$, the set $\{ar_1, \dots, ar_{\varphi(m)}\}$ is also a reduced residue system modulo m (by problem 1 in next section). Thus,

$$r_1 r_2 \cdots r_{\varphi(m)} \equiv (ar_1)(ar_2) \cdots (ar_{\varphi(m)}) \equiv a^{\varphi(m)} (r_1 r_2 \cdots r_{\varphi(m)}) \pmod{m}.$$

Since $\gcd(r_1 r_2 \cdots r_{\varphi(m)}, m) = 1$, we can cancel it from both sides to get

$$a^{\varphi(m)} \equiv 1 \pmod{m}.$$

■

2. Let $m > 1$ and let $\{r_1, \dots, r_{\varphi(m)}\}$ be a reduced residue system modulo m . Characterize integers j for which $\{jr_1, \dots, jr_{\varphi(m)}\}$ is also a reduced residue system modulo m .

Proof: If $\gcd(j, m) = 1$ then multiplication by j is a bijection on $(\mathbb{Z}/m\mathbb{Z})^\times$, so the set $\{jr_i\}$ consists of distinct residues all coprime to m , hence is a reduced residue system. Conversely, if $\gcd(j, m) > 1$ then any r_i with $\gcd(r_i, m) = 1$ yields $\gcd(jr_i, m) > 1$, so the image contains nonunits and cannot be a reduced residue system.

■

3. If $a, b \in \mathbb{Z}$, $a \geq b > 0$, and $a = bq + r$ (with $0 \leq r < b$), show $(a, b) = (b, r)$.

Proof: Let $d = (a, b)$. Then $d \mid a$ and $d \mid b$, so $d \mid (a - bq) = r$. Thus, $d \mid b$ and $d \mid r$, so $d \leq (b, r)$. Conversely, let $e = (b, r)$. Then $e \mid b$ and $e \mid r$, so $e \mid (bq + r) = a$. Thus, $e \mid a$ and $e \mid b$, so $e \leq (a, b)$. Therefore, $(a, b) = (b, r)$.

■

4. If $a, b \in \mathbb{Z}$ and $(a, b) = [a, b]$, what can you say about a, b ?

Proof: If $(a, b) = [a, b]$ then a and b are coprime, i.e., $\gcd(a, b) = 1$.

■

5. Let $a, b \in \mathbb{Z}$ and p prime. If $p \mid ab$ then $p \mid a$ or $p \mid b$.

Proof: If $p \nmid a$ then $\gcd(a, p) = 1$, so there exist integers x, y such that $ax + py = 1$. Multiplying by b gives $abx + pby = b$. Since $p \mid ab$ and $p \mid pby$, we have $p \mid b$.

■

6. Let p be prime and $p \nmid a$. Show there is a unique solution to $ax \equiv 1 \pmod{p}$.

Proof: Since $\gcd(a, p) = 1$, a is a unit in $\mathbb{Z}/p\mathbb{Z}$ so an inverse x exists. If x_1, x_2 are both inverses then $a(x_1 - x_2) \equiv 0 \pmod{p}$. As $p \nmid a$, we get $p \mid (x_1 - x_2)$, so $x_1 \equiv x_2 \pmod{p}$. Thus the inverse is unique modulo p .

■

7. Let $a, b, c, m \in \mathbb{Z}$ with $m, c \geq 1$. If $a \equiv b \pmod{m}$ then $ac \equiv bc \pmod{mc}$.

Proof: $a \equiv b \pmod{m}$ implies $m \mid (a - b)$, so $mc \mid c(a - b) = ac - bc$. Therefore $ac \equiv bc \pmod{mc}$. ■

8. Let $a, b, c, d \in \mathbb{Z}$ with $b, d \neq 0$ and $(a, b) = (c, d) = 1$. If $\frac{a}{b} + \frac{c}{d}$ is an integer, show $b = \pm d$.

Proof: Suppose $\frac{a}{b} + \frac{c}{d} = k \in \mathbb{Z}$. Clearing denominators gives

$$ad + bc = kbd \quad \Rightarrow \quad ad = b(kd - c).$$

Thus $b \mid ad$. Since $\gcd(a, b) = 1$, we conclude $b \mid d$. Symmetrically $d \mid b$. Hence $|b| = |d|$, so $b = \pm d$. ■